

1. Algorithms, Approaches, and Tools

1.1 Principal Component Analysis (PCA)

Principal Component Analysis is a linear dimensionality-reduction method that transforms a high-dimensional dataset into a new set of orthogonal variables called principal components. The first principal component, PC1, captures the greatest possible variance in the data, while the second principal component, PC2, captures the greatest remaining variance while being orthogonal to PC1.

PCA accomplishes this by centering the feature data and finding directions in the original feature space in which the samples vary the most. The original 12,309-dimensional observations are projected onto the two leading directions, producing one PC1 and PC2 values for every sample.

In task 1, PCA was applied to the lncRNA expression features, and the first two components were plotted as a two-dimensional scatter plot. In Task 2, the PC1 and PC2 values were separately visualized with violin plots to compare their distributions across the five cancer classes.

1.2 t-Distributed Stochastic Neighbor Embedding (t-SNE)

t-SNE is a nonlinear dimensionality-reduction technique designed primarily for visualization. Unlike PCA, which is used for emphasizing global variance, t-SNE emphasizes preservation of local neighborhoods so that samples that are similar in the original high-dimensional space tend to remain near one another in the two-dimensional visualization.

It achieves this by converting pairwise similarities among high-dimensional samples into probability-like neighborhood relationships and then searches for a two-dimensional arrangement with similar neighborhood relationships. Because the optimization is nonlinear and can depend on initialization and parameter choices, the absolute position, and scale of the final t-SNE axes are not fundamentally meaningful.

In task 3, the lncRNA samples were mapped to two t-SNE coordinates and visualized by cancer type. In task 4, violin plots were used to show the distributions of t-SNE 1 and t-SNE 2 for each class.

1.3 Visualization and Software Tools

Python was used to perform the analysis. The pandas library was used to load and organize the CSV data; scikit-learn supplied the PCA and t-SNE implementations, and

Matplotlib/Seaborn were used to construct scatter plots and violin plots. Scatter plots show the two-dimensional location of every sample, and violin plots combine a density estimate with summary statistics to show where values are concentrated and how widely they are distributed.

2. Results and Observations

2.1 Task 1 – PCA Visualization

Figure 1 shows the first two PCA components for all five cancer types. The projection reveals substantial class structure even after reducing 12,309 expression measurements to only two coordinates. KIRC is concentrated primarily in the upper-left portion of the plot, PRAD forms a distinct group on the far right, and THCA extends diagonally through the central/upper-right region. LUAD and LUSC occupy the lower central region and show noticeably more overlap with each other than with KIRC or PRAD. This indicates that the largest linear sources of variation in the expression data contain useful information for distinguishing the cancer types, although two PCA dimensions do not completely separate every class.

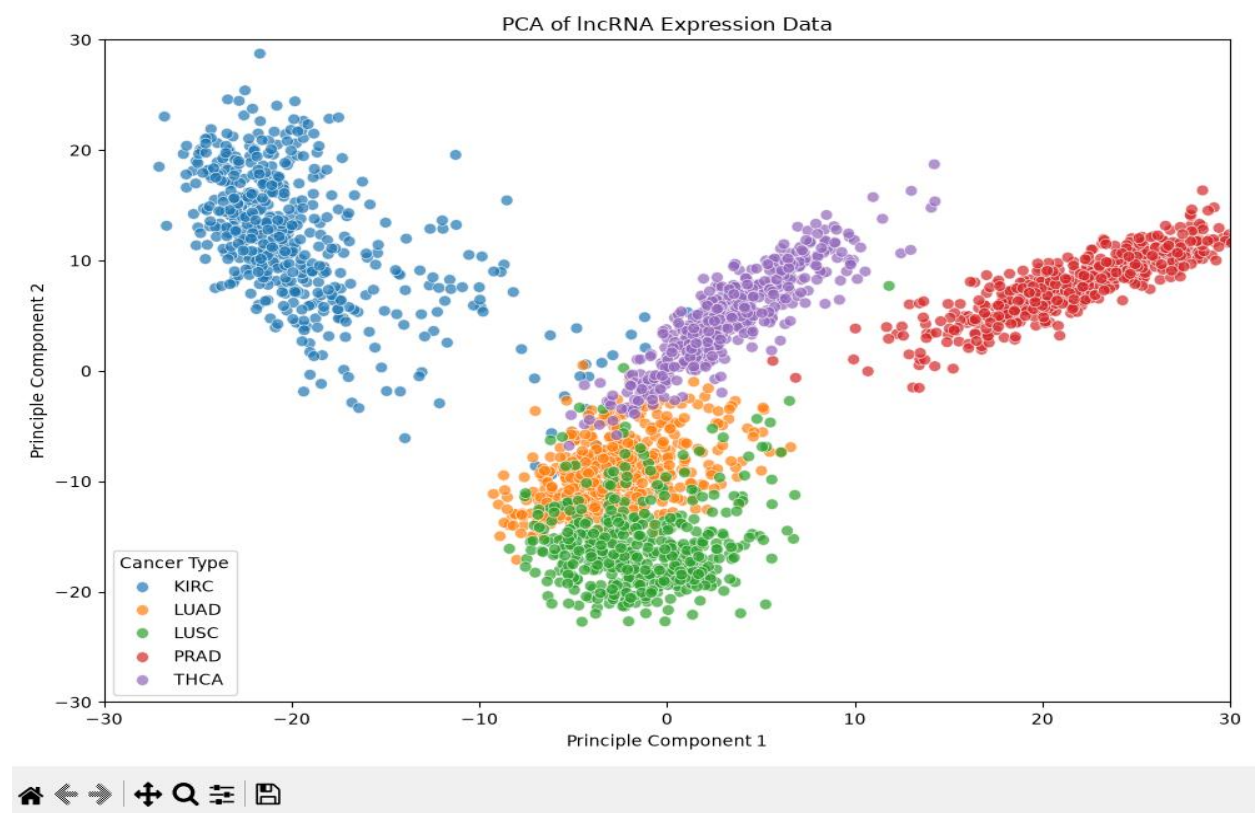


Figure 1. PCA visualization of lncRNA expression data for five cancer types

3.2 Task 2 – PCA Violin Plots

Figures 2 and 3 provide one-dimensional views of the same PCA embedding. In Figure 2, KIRC has strongly negative PC1 values, PRAD has strongly positive PC1 values, and THCA is moderately positive. LUAD and LUSC are both centered at slightly negative PC1 values, which is consistent with their closer positioning in the PCA scatter plot. The separation between KIRC and PRAD along PC1 shows that the first component is particularly informative for distinguishing these classes.

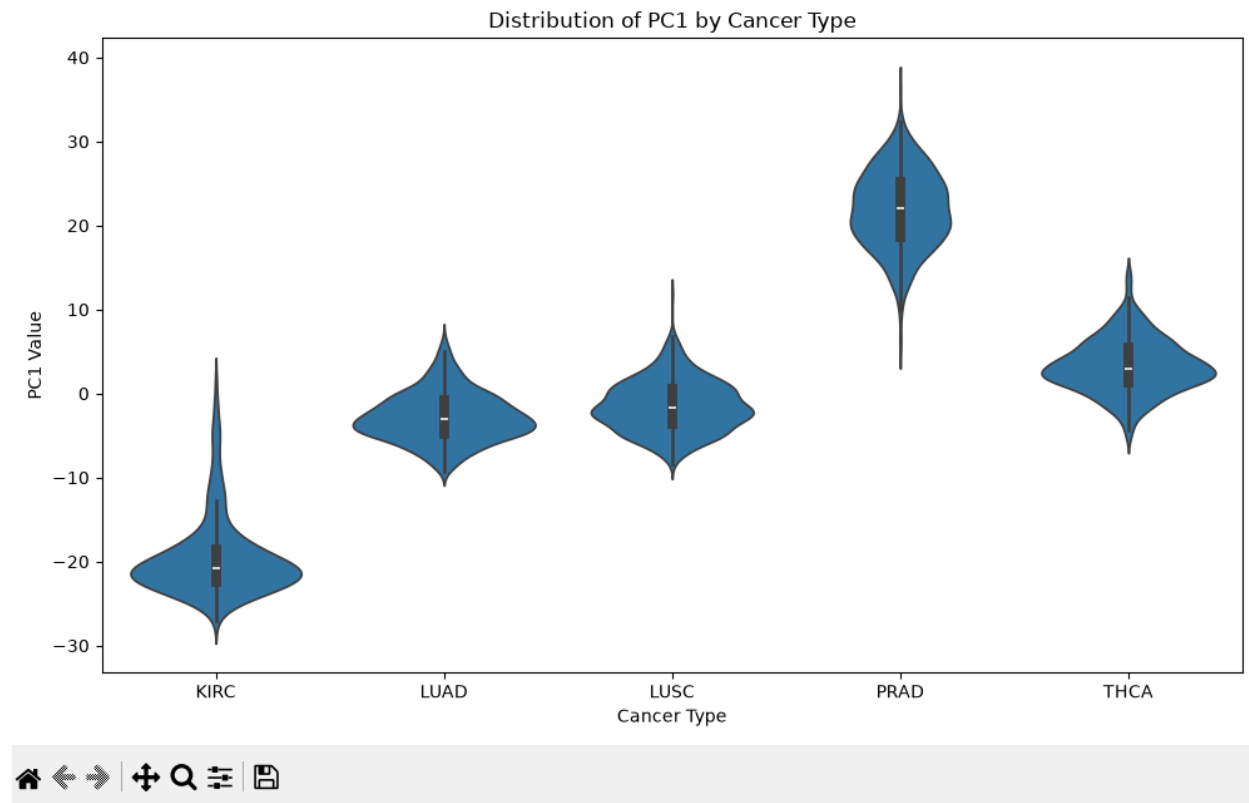


Figure 2. Violin plot showing the distribution of PC1 values by cancer type.

Figure 3 shows that the PC2 provides a different pattern of separation. KIRC is concentrated at positive PC2 values, while LUAD and especially LUSC are concentrated at negative values. PRAD and THCA are generally positive on PC2. Together, the PC1 and PC2 violin plots explain why the cancer groups occupy different regions of Figure 1: each class has a characteristic distribution across the two principal-component coordinates.

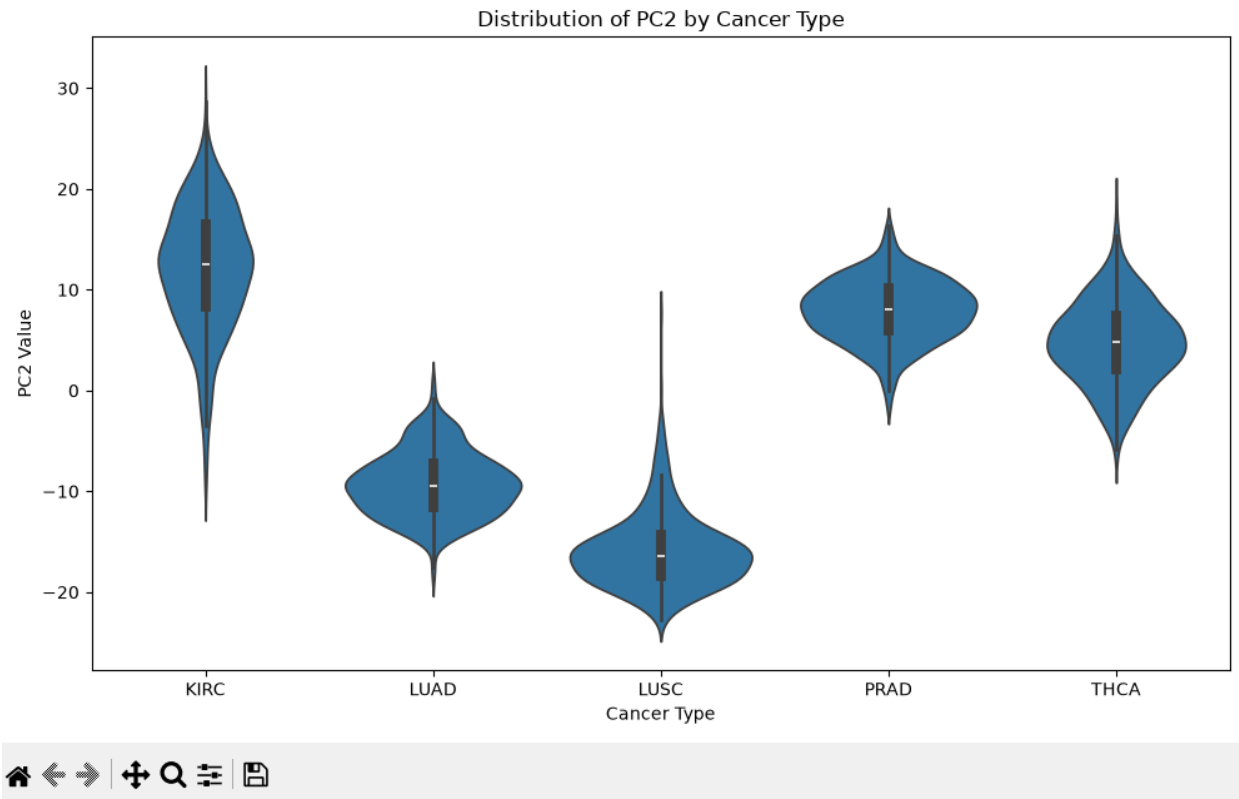


Figure 3. Violin plot showing the distribution of PC2 values by cancer type.

2.3 Task 3 – t-SNE Visualization

Figure 4 shows the two-dimensional t-SNE embedding. Compared with PCA, t-SNE produces more visibly separated local groups for several cancer types. KIRC forms a cluster on the left, THCA is concentrated in the upper central region, and PRAD forms a clearly separated cluster on the right. LUSC is concentrated in the lower-left/central region. LUAD lies primarily in the lower central region and has the most visible local mixing with LUSC. The stronger visual grouping is consistent with t-SNE's objective of preserving local neighborhoods rather than maximizing global linear variance.

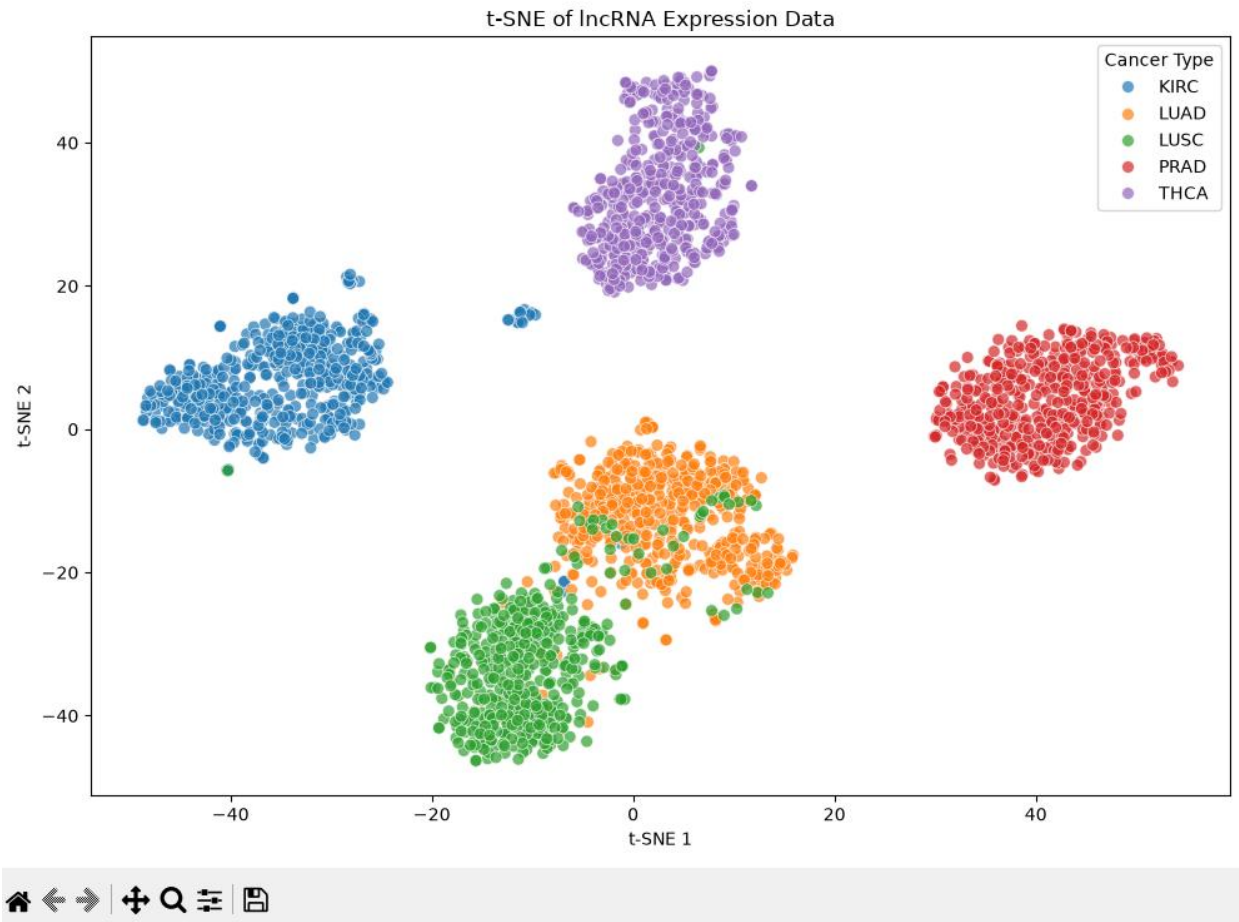


Figure 4. t-SNE visualization of lncRNA expression data for five cancer types.

The numerical t-SNE coordinates should not be interpreted as biological quantities. For example, the fact that PRAD appears on the right side of the graph is not itself meaningful. A different valid t-SNE run could rotate, reflect, translate, or rescale the embedding. The meaningful observation is the neighborhood and clustering structure.

3.4 Task – t-SNE Violin Plots

Figure 5 summarizes the distribution of the first t-SNE coordinate. KIRC is concentrated at large negative t-SNE 1 values, PRAD at large positive values, LUAD near zero, and THCA also near zero. LUSC is primarily negative. These distributions correspond to the left-to-right locations of the clusters in Figure 4.

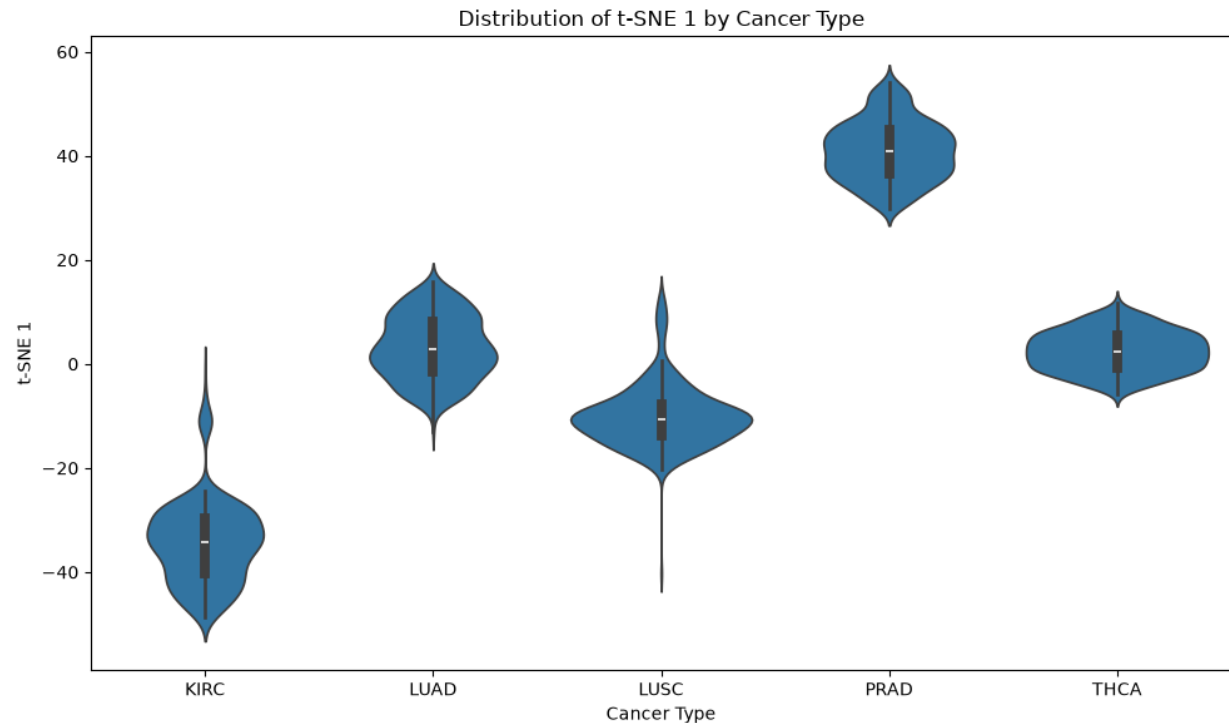


Figure 5. Violin plot showing the distribution of t-SNE 1 values by cancer type.

Figure 6 summarizes the second t-SNE coordinate. THCA is concentrated at high positive t-SNE 2 values, whereas LUSC is concentrated at strongly negative values. LUAD is also mostly negative, while KIRC and PRAD are centered around small positive values. These distributions correspond to the vertical placement of the groups in the t-SNE scatter plot.

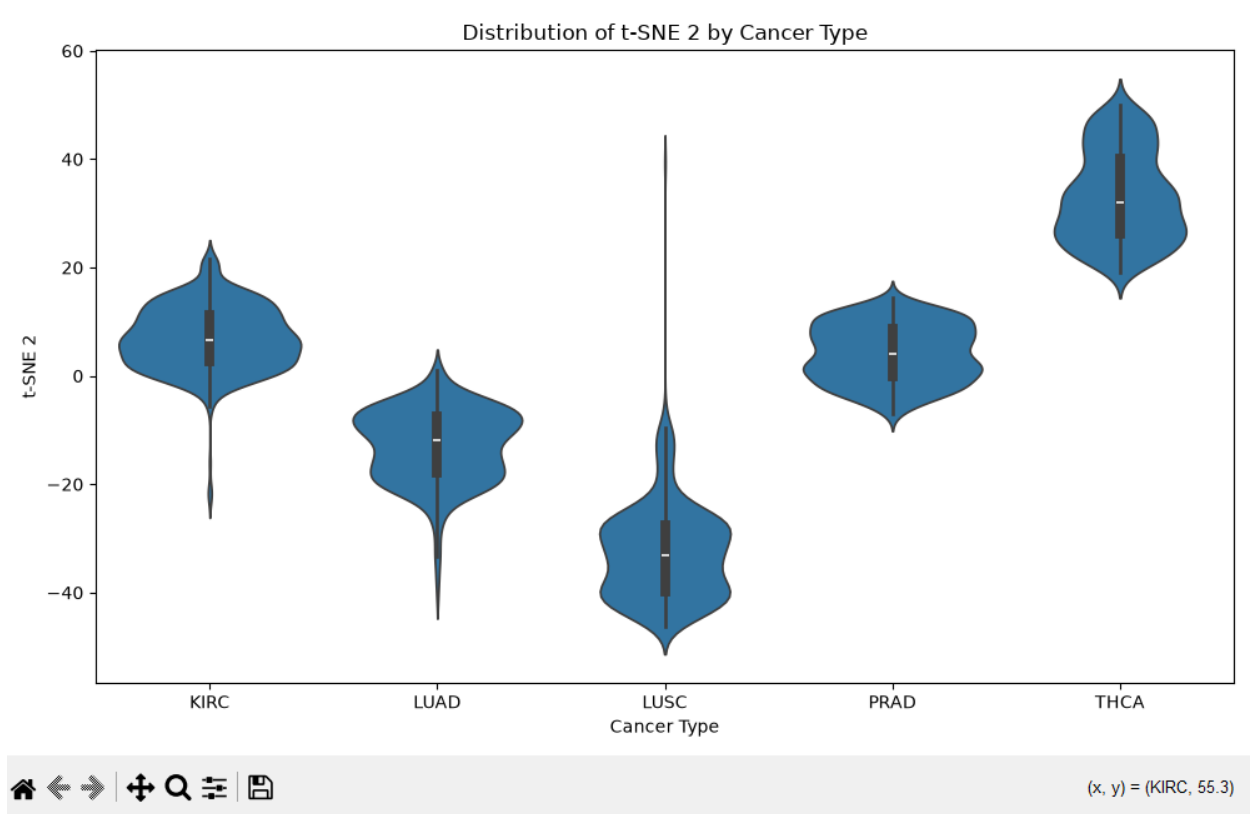


Figure 6. Violin plot shows the distribution of t-SNE 2 values by cancer type.

3. Conclusion

This assignment demonstrated how dimensionality-reduction techniques make very high-dimensional biological data easier to explore visually. PCA reduced the 12,309 lncRNA expression dimensions to PC1 and PC2 while preserving major linear variation, revealing clear separation for several cancer types and some overlap between LUAD and LUSC. T-SNE provided an alternative nonlinear representation with stronger visual clustering of several classes while still showing some LUAD/LUSC proximity. The violin plots complemented the scatter plots by showing the distributions of each reduced coordinate for every cancer type. Overall, the results demonstrate that both PCA and t-SNE can expose meaningful structure in high-dimensional gene-expression data, while also illustrating the different kinds of structure each method is designed to preserve.